Meta-Learning with Reptile for Efficient Few-Shot Adaptation

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Topic

This project focuses on *meta-learning*, a paradigm in machine learning where models are trained to rapidly adapt to new tasks using limited data. In particular, we investigate gradient-based meta-learning methods for few-shot learning.

Project Description

The objective of this project is to implement and evaluate the Reptile algorithm, a first-order meta-learning method proposed by Nichol and Schulman (2018). Unlike conventional supervised learning approaches that require large amounts of task-specific data, meta-learning aims to learn a shared initialization that can be efficiently adapted to new tasks with only a few gradient updates.

In this project, we will develop a complete pipeline for training and evaluating the Reptile algorithm. The implementation will first be validated on a controlled regression problem involving sine wave fitting, where each task corresponds to a function with varying amplitude and phase. This setting allows for clear visualization and debugging of the meta-learning process.

Subsequently, we will extend the approach to few-shot image classification tasks using benchmark datasets such as Omniglot, and optionally Mini-ImageNet depending on computational constraints. The goal is to assess how effectively the learned initialization enables rapid adaptation in low-data regimes.

This project is directly relevant to the course as it explores advanced learning frameworks that go beyond standard supervised learning, emphasizing generalization across tasks and efficient learning from limited data.

Datasets

We use a synthetic sine wave dataset to validate the implementation of the Reptile algorithm in a controlled setting. Each task corresponds to a sine function of the form $f(x) = a \sin(x + b)$, where the amplitude and phase are randomly sampled. For each task, a small set of input-output pairs is generated and split into a support set for adaptation and a query set for evaluation. This dataset allows us to easily analyze whether the model can learn a good initialization for rapid adaptation across tasks.

The Omniglot dataset is used for few-shot image classification and consists of handwritten characters from multiple alphabets. Each character is treated as a separate class, while the alphabet structure is ignored. The dataset is divided into *images_background* for training and *images_evaluation* for testing, ensuring that the model is evaluated on unseen characters. Tasks are constructed in an episodic manner by sampling a subset of classes and forming support and query sets.

To evaluate performance on more complex visual data, we use the Mini-ImageNet dataset, which contains natural images from 100 object classes. The dataset is split into training, validation, and test classes, and tasks are generated following the standard N-way K-shot setting. Compared to Omniglot, Mini-ImageNet introduces greater variability and complexity, making it suitable for assessing the scalability of the meta-learning approach.

Methodology

The project will be implemented using PyTorch, which provides flexibility for defining custom training loops required in meta-learning.

The methodology consists of two nested optimization loops operating over a distribution of tasks. In each iteration, a task is sampled and the model parameters are adapted using a small number of gradient descent steps (inner loop). Specifically, the model is initialized with parameters ϕ, which are copied to task-specific parameters W and updated using the support set.

In the outer loop, the initialization parameters are updated towards the task-specific parameters obtained after adaptation, following the Reptile update rule:

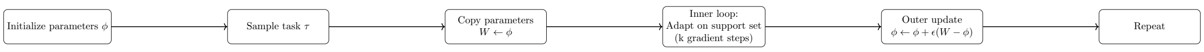
$$\phi \leftarrow \phi + \epsilon(W - \phi),$$

where ϵ is the meta learning rate.

We will begin with a regression setup where tasks are generated from a distribution of sine functions. Each task involves sampling input-output pairs and adapting the model to fit the function. This controlled setup allows us to verify correctness and study the effect of hyperparameters such as the number of inner-loop steps and learning rates.

For classification experiments, we adopt the standard N-way K-shot setting. Tasks are constructed by sampling classes and examples from the dataset, forming support and query sets. The model is trained to learn an initialization that enables rapid adaptation to new classification tasks with only a few labeled examples.

We will also compare the performance of the proposed approach with baseline methods such as standard supervised training and simple fine-tuning from random initialization. If time permits, a comparison with first-order MAML will also be included.



Evaluation

In contrast to standard supervised learning, where models are evaluated directly on a fixed test set, meta-learning requires an episodic evaluation protocol. In this project, performance is assessed after adapting the model to a new task using a small number of gradient steps. For each test task, the model is initialized with the learned parameters, adapted using the support set, and then evaluated on the corresponding query set.

For regression tasks (sine wave dataset), performance is measured using mean squared error (MSE). For classification tasks (Omniglot and Mini-ImageNet), performance is evaluated using classification accuracy. Evaluation is performed over a large number of sampled tasks from the test split, and results are reported as the average performance along with the standard deviation across tasks.

In addition to aggregate metrics, we analyze the adaptation behavior by measuring performance as a function of the number of inner-loop gradient steps. This allows us to observe how quickly the model improves with additional adaptation on new tasks.

5. Resources

The project will utilize Python and PyTorch as the primary development framework, along with NumPy for numerical operations and Matplotlib for visualization.

The datasets used will include a synthetically generated sine wave dataset for regression experiments and the Omniglot dataset for few-shot classification. Mini-ImageNet may be used as an additional benchmark if computational resources allow.

The primary reference for this project is the paper *"Reptile: A Scalable Meta-Learning Algorithm"* by Nichol and Schulman (2018). We will also refer to *"Model-Agnostic Meta-Learning (MAML)"* by Finn et al. (2017) for conceptual comparison. Publicly available GitHub repositories will be consulted for reference implementations and best practices, but the core implementation will be developed independently.

Work Distribution

- Shaikh Asif Hossain will focus on implementing the Reptile algorithm, including the inner and outer optimization loops, as well as designing the training pipeline.
- Sorana Resiga will be responsible for dataset preparation, experimental setup, evaluation, and analysis of results.

Expected Outcomes

The expected outcome of this project is a complete and well-documented implementation of the Reptile meta-learning algorithm, along with empirical results demonstrating its ability to adapt quickly to new tasks. We expect to gain insights into the effectiveness of meta-learning in low-data settings and understand the trade-offs between simplicity and performance compared to other methods such as MAML.

References

[1] A. Nichol and J. Schulman, *Reptile: A Scalable Meta-Learning Algorithm*, 2018.

[2] C. Finn, P. Abbeel, and S. Levine, *Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks*, Proceedings of the 34th International Conference on Machine Learning (ICML), 2017.

[3] B. Lake, R. Salakhutdinov, and J. Tenenbaum, *Human-level concept learning through probabilistic program induction*, Science, 2015.

[4] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, *ImageNet: A Large-Scale Hierarchical Image Database*, CVPR, 2009.